Since clearly it is also true that

$$\sum_j P_j = 1$$

it thus follows, since by Theorem 2.1 the $\Pi_j, j = 0, \ldots, M$ are the unique solution of the preceding, that $P_j = \Pi_j, j = 0, \ldots, M$. The longrun proportion interpretation of Π_j, is generally valid even when the chain is not ergodic.

Example 2f. Suppose in Example 2c that we are interested in the proportion of time there are j molecules in urn 1, $j = 0, \ldots, M$. By Theorem 2.1 these quantities will be the unique solution of

$$\Pi_0 = \Pi_1 \times \frac{1}{M}$$

$$\Pi_j = \Pi_{j-1} \times \frac{M - j + 1}{M} + \Pi_{j+1} \times \frac{j + 1}{M} \qquad j = 1, \ldots, M$$

$$\Pi_M = \Pi_{M-1} \times \frac{1}{M}$$

$$\sum_{j=0}^{M} \Pi_j = 1$$

However, as it is easily checked that

$$\Pi_j = \binom{M}{j}\left(\frac{1}{2}\right)^M \qquad j = 0, \ldots, M$$

satisfy the preceding equations, it follows that these are the long-run proportions of time that the Markov chain is in each of the states. (See Problem 11 for an explanation of how one might have guessed at the foregoing solution.) ■

9.3 SURPRISE, UNCERTAINTY, AND ENTROPY

Consider an event E that can occur when an experiment is performed. How surprised would we be to hear that E does, in fact, occur? It seems reasonable to suppose that the amount of surprise engendered by the information that E has occurred should depend on the probability of E. For instance, if the experiment consists of rolling a pair of dice, then we would not be too surprised to hear that E has occurred when E represents the event that the sum of the dice is even (and thus has probability $\frac{1}{2}$), whereas we would certainly be more surprised to hear that E has occurred when E is the event that the sum of the dice is 12 (and thus has probability $\frac{1}{36}$).

In this section we attempt to quantify the concept of surprise. To begin, let us agree to suppose that the surprise one feels upon learning that an event E has occurred depends only on the probability of E; and let us denote by $S(p)$ the surprise evoked by the occurrence of an event having probability p. We determine the functional form of $S(p)$ by first agreeing on a set of reasonable conditions that

$S(p)$ should satisfy, and then proving that these axioms require that $S(p)$ has a specified form. We assume throughout that $S(p)$ is defined for all $0 < p \leq 1$ but is not defined for events having $p = 0$.

Our first condition is just a statement of the intuitive fact that there is no surprise in hearing that an event sure to occur has indeed occurred.

Axiom 1

$$S(1) = 0$$

Our second condition states that the more unlikely an event is to occur, the greater is the surprise evoked by its occurrence.

Axiom 2

$S(p)$ is a strictly decreasing function of p; that is, if $p < q$, then $S(p) > S(q)$.

The third condition is a mathematical statement of the fact that we would intuitively expect a small change in p to correspond to a small change in $S(p)$.

Axiom 3

$S(p)$ is a continuous function of p.

To motivate the final condition, consider two independent events E and F, having respective probabilities $P(E) = p$ and $P(F) = q$. Since $P(EF) = pq$, the surprise evoked by the information that both E and F have occurred is $S(pq)$. Now, suppose that we are first told that E has occurred and then, afterward, that F has also occurred. As $S(p)$ is the surprise evoked by the occurrence of E, it follows that $S(pq) - S(p)$ represents the additional surprise evoked when we are informed that F has also occurred. However, as F is independent of E, the knowledge that E occurred does not change the probability of F, and hence the additional surprise should just be $S(q)$. This reasoning suggests the final condition.

Axiom 4

$$S(pq) = S(p) + S(q) \qquad 0 < p \leq 1, 0 < q \leq 1$$

We are now ready for Theorem 3.1, which yields the structure of $S(p)$.

Theorem 3.1

If $S(\cdot)$ satisfies Axioms 1 through 4, then

$$S(p) = -C \log_2 p$$

where C is an arbitrary positive integer.

Proof: It follows from Axiom 4 that

$$S(p^2) = S(p) + S(p) = 2S(p)$$

and by induction that

$$S(p^m) = m\,S(p) \tag{3.1}$$

Also, since for any integral n, $S(p) = S(p^{1/n} \cdots p^{1/n}) = n\,S(p^{1/n})$, it follows that

$$S(p^{1/n}) = \frac{1}{n} S(p) \tag{3.2}$$

Thus, from Equations (3.1) and (3.2), we obtain

$$\begin{aligned} S(p^{m/n}) &= m\,S(p^{1/n}) \\ &= \frac{m}{n} S(p) \end{aligned}$$

which is equivalent to

$$S(p^x) = xS(p) \tag{3.3}$$

whenever x is a positive rational number. But this implies by the continuity of S (Axiom 3) that Equation (3.3) is valid for all nonnegative x. (Reason this out.)

Now, for any p, $0 < p \le 1$, let $x = -\log_2 p$. Then $p = \left(\frac{1}{2}\right)^x$ and from Equation (3.3),

$$S(p) = S\left(\left(\tfrac{1}{2}\right)^x\right) = xS\left(\tfrac{1}{2}\right) = -C \log_2 p$$

where $C = S\left(\frac{1}{2}\right) > S(1) = 0$ by Axioms 2 and 1. ✦

It is usual to let C equal 1. In this case the surprise is said to be expressed in units of *bits* (short for *binary digits*).

Consider now a random variable X, which must take on one of the values $x_1, \ldots, x_n$ with respective probabilities $p_1, \ldots, p_n$. As $-\log p_i$ represents the surprise evoked if X takes on the value x_i,[†] it follows that the expected amount of surprise we shall receive upon learning the value of X is given by

$$H(X) = -\sum_{i=1}^{n} p_i \log p_i$$

[†] For the remainder of this chapter we write $\log x$ for $\log_2 x$. Also, we use $\ln x$ for $\log_e x$.

The quantity $H(X)$ is known in information theory as the *entropy* of the random variable X. (In case one of the $p_i = 0$, we take $0 \log 0$ to equal 0.) It can be shown (and we leave it as an exercise) that $H(X)$ is maximized when all of the p_i are equal. (Is this intuitive?)

As $H(X)$ represents the average amount of surprise one receives upon learning the value of X, it can also be interpreted as representing the amount of *uncertainty* that exists as to the value of X. In fact, in information theory, $H(X)$ is interpreted as the average amount of *information* received when the value of X is observed. Thus the average surprise evoked by X, the uncertainty of X, or the average amount of information yielded by X, all represent the same concept viewed from three slightly different points of view.

Consider now two random variables, X and Y, which take on respective values $x_1, \ldots, x_n$ and $y_1, \ldots, y_m$ with joint mass function

$$p(x_i, y_j) = P\{X = x_i, Y = y_j\}$$

It follows that the uncertainty as to the value of the random vector (X, Y), denoted by $H(X, Y)$, is given by

$$H(X, Y) = -\sum_i \sum_j p(x_i, y_j) \log p(x_i, y_j)$$

Suppose now that Y is observed to equal y_j. In this situation the amount of uncertainty remaining in X is given by

$$H_{Y=y_j}(X) = -\sum_i p(x_i|y_j) \log p(x_i|y_j)$$

where

$$p(x_i|y_j) = P\{X = x_i|Y = y_j\}$$

Hence the average amount of uncertainty that will remain in X after Y is observed is given by

$$H_Y(X) = \sum_j H_{Y=y_j}(X) p_Y(y_j)$$

where

$$p_Y(y_j) = P\{Y = y_j\}$$

Proposition 3.1 relates $H(X, Y)$ to $H(Y)$ and $H_Y(X)$. It states that the uncertainty as to the value of X and Y is equal to the uncertainty of Y plus the average uncertainty remaining in X when Y is to be observed.

Proposition 3.1

$$H(X, Y) = H(Y) + H_Y(X)$$

Proof: Using the identity $p(x_i, y_j) = p_Y(y_j)p(x_i|y_j)$ yields

$$\begin{aligned} H(X,Y) &= -\sum_i \sum_j p(x_i, y_j) \log p(x_i, y_j) \\ &= -\sum_i \sum_j p_Y(y_j)p(x_i|y_j)[\log p_Y(y_j) + \log p(x_i|y_j)] \\ &= -\sum_j p_Y(y_j) \log p_Y(y_j) \sum_i p(x_i|y_j) \\ &\quad -\sum_j p_Y(y_j) \sum_i p(x_i|y_j) \log p(x_i|y_j) \\ &= H(Y) + H_Y(X) \end{aligned}$$

✦

It is a fundamental result in information theory that the amount of uncertainty in a random variable X will, on the average, decrease when a second random variable Y is observed. Before proving this, we need the following lemma, whose proof is left as an exercise.

Lemma 3.1

$$\ln x \le x - 1 \qquad x > 0$$

with equality only at $x = 1$.

Theorem 3.2

$$H_Y(X) \le H(X)$$

with equality if and only if X and Y are independent.

Proof:

$$\begin{aligned} H_Y(X) - H(X) &= -\sum_i \sum_j p(x_i|y_j) \log[p(x_i|y_j)]p(y_j) \\ &\quad + \sum_i \sum_j p(x_i, y_j) \log p(x_i) \\ &= \sum_i \sum_j p(x_i, y_j) \log\left[\frac{p(x_i)}{p(x_i|y_j)}\right] \\ &\le \log e \sum_i \sum_j p(x_i, y_j)\left[\frac{p(x_i)}{p(x_i|y_j)} - 1\right] \qquad \text{by Lemma 3.1} \end{aligned}$$

$$= \log e\left[\sum_i \sum_j p(x_i)p(y_j) - \sum_i \sum_j p(x_i, y_j)\right]$$
$$= \log e[1 - 1]$$
$$= 0$$

✦

9.4 CODING THEORY AND ENTROPY

Suppose that the value of a discrete random vector X is to be observed at location A and then transmitted to location B via a communication network that consists of two signals, 0 and 1. In order to do this, it is first necessary to encode each possible value of X in terms of a sequence of 0's and 1's. To avoid any ambiguity, it is usually required that no encoded sequence can be obtained from a shorter encoded sequence by adding more terms to the shorter.

For instance, if X can take on four possible values x_1, x_2, x_3, x_4, then one possible coding would be

$$\begin{aligned} x_1 &\leftrightarrow 00 \\ x_2 &\leftrightarrow 01 \\ x_3 &\leftrightarrow 10 \\ x_4 &\leftrightarrow 11 \end{aligned} \tag{4.1}$$

That is, if $X = x_1$, then the message 00 is sent to location B, whereas if $X = x_2$, then 01 is sent to B, and so on. A second possible coding is

$$\begin{aligned} x_1 &\leftrightarrow 0 \\ x_2 &\leftrightarrow 10 \\ x_3 &\leftrightarrow 110 \\ x_4 &\leftrightarrow 111 \end{aligned} \tag{4.2}$$

However, a coding such as

$$\begin{aligned} x_1 &\leftrightarrow 0 \\ x_2 &\leftrightarrow 1 \\ x_3 &\leftrightarrow 00 \\ x_4 &\leftrightarrow 01 \end{aligned}$$

is not allowed because the coded sequences for x_3 and x_4 are both extensions of the one for x_1.

One of the objectives in devising a code is to minimize the expected number of bits (that is, binary digits) that need be sent from location A to location B. For example, if